The maximum entry in reverse plane partitions and cylindric plane partitions

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Abstract

We collect what is known about reverse plane partitions (RPPs) and cylindric plane partitions (CPPs) refined by their maximum entry. The maximum entry n governs the "depth" of the combinatorial object: it is the number of layers in the level-set decomposition, the number of non-intersecting lattice paths in the NILP model, and the number of insertion steps in the RSK/growth-diagram framework. We describe the transfer matrix formulation (products of skew Schur functions), Borodin's product formula, the E–H factorization that extracts Warnaar's $Q_{n,c}(q)$ polynomials, and the partial quasi-symmetry of the frequency refinement at fixed maximum entry.

Contents

1 Reverse plane partitions and the maximum entry

1.1 Definitions

A *reverse plane partition* (RPP) of shape λ is a filling of the Young diagram of λ with nonneg integers that is weakly increasing along rows (left to right) and weakly increasing down columns (top to bottom). We write $\mathrm{RPP}(\lambda)$ for the set of all RPPs of shape λ.

Remark 1. *Some authors use the convention that entries are weakly* decreasing *("plane partitions" in the sense of MacMahon). We follow the reverse/increasing convention, which matches the interlacing-sequence model and Stanley's P-partition framework.*

The *weight* of an RPP π is $|\pi| = \sum_{(i,j)\in\lambda} \pi_{ij}$, and the *maximum entry* is $\max(\pi) = \max_{(i,j)} \pi_{ij}$.

Stanley's celebrated formula [**?**] gives the weight generating function:

$$\sum_{\pi\in \mathrm{RPP}(\lambda)} q^{|\pi|} = \prod_{u\in\lambda} \frac{1}{1-q^{h_\lambda(u)}}, \tag{1}$$

where $h_\lambda(u)$ is the hook length at box u.

1.2 Level-set decomposition

Given an RPP π with $\max(\pi) = n$, define the *level sets*:

$$L_k(\pi) = \{(i,j) \in \lambda : \pi_{ij} \geq k\}, \qquad k = 1, 2, \ldots, n.$$

Since π is weakly increasing in both directions, each $L_k(\pi)$ is a Young diagram (the upper-left corner condition is preserved). We obtain a chain of partitions:

$$\lambda = L_0 \supseteq L_1 \supseteq L_2 \supseteq \cdots \supseteq L_n \supseteq \emptyset, \tag{2}$$

and the weight decomposes as $|\pi| = \sum_{k=1}^{n} |L_k|$ (each box at height $\geq k$ contributes 1 for each level).

The RPP is uniquely determined by its level-set chain. Thus:

Proposition 2. *RPPs of shape λ with* $\max \leq n$ *are in bijection with chains of partitions $\lambda \supseteq \mu^{(1)} \supseteq \cdots \supseteq \mu^{(n)} \supseteq \emptyset$ inside λ.*

This is the "n-layer" decomposition: max entry $= n$ means exactly n layers.

1.3 The interlacing-sequence model

Reading an RPP along its anti-diagonals gives an *interlacing sequence* of partitions. For shape λ with profile (binary string) $\pi = (\pi_1, \ldots, \pi_T)$:

$$\emptyset = \nu^{(0)},\ \nu^{(1)},\ \ldots,\ \nu^{(T)},\ \nu^{(T+1)} = \emptyset,$$

where $\nu^{(k-1)}$ and $\nu^{(k)}$ satisfy the interlacing condition determined by π_k:

- If $\pi_k = 0$: $\nu^{(k)}/\nu^{(k-1)}$ is a horizontal strip (row added);
- If $\pi_k = 1$: $\nu^{(k-1)}/\nu^{(k)}$ is a horizontal strip (row removed).

In this model, the maximum entry of the RPP equals $\max_{k,j} \nu_j^{(k)}$, the maximum part across all partitions in the sequence.

1.4 Bounding the maximum entry

For RPPs of shape λ with max entry $\leq m$, we restrict all parts $\nu_j^{(k)} \leq m$ in the interlacing sequence. This is equivalent to requiring that each partition $\nu^{(k)}$ has at most m columns (in English convention), or equivalently that the conjugate $(\nu^{(k)})'$ has at most m parts.

The generating function for RPPs of shape λ refined by maximum entry is:

$$\sum_{\pi \in \mathrm{RPP}(\lambda)} z^{\max(\pi)} q^{|\pi|} = \sum_{n \geq 0} g_{n,\lambda}(q)\, z^n. \tag{3}$$

The variable z tracks the maximum entry, and $g_{n,\lambda}(q)$ counts RPPs of shape λ with exact max entry n by weight.

2 Cylindric plane partitions

2.1 Definition and generating function

A *rank-k cylindric plane partition* with profile $c = (c_1, \ldots, c_k)$ is a k-tuple of partitions $(\nu^{(1)}, \ldots, \nu^{(k)})$ satisfying the *cyclic* interlacing conditions:

$$\nu_j^{(i)} \geq \nu_{j+c_{i+1}}^{(i+1)}, \qquad i = 1, \ldots, k, \quad \text{indices mod } k. \tag{4}$$

When $k = 3$ and $c = (c_0, c_1, c_2)$ with $d = |c|$:

$$\nu_j^{(1)} \geq \nu_{j+c_1}^{(2)}, \qquad \nu_j^{(2)} \geq \nu_{j+c_2}^{(3)}, \qquad \nu_j^{(3)} \geq \nu_{j+c_0}^{(1)}.$$

RPPs are the special case where the profile corresponds to a partition shape and the boundary condition is $\nu^{(0)} = \nu^{(T+1)} = \emptyset$ (non-cyclic). The cyclic boundary condition is what distinguishes CPPs.

The two-variable generating function is:

$$\mathrm{GK}_c(z, q) = \sum_{\pi \in \mathrm{CPP}(c)} z^{n(\pi)} q^{|\pi|} = \sum_{n \geq 0} g_n(q)\, z^n, \tag{5}$$

where $n(\pi) = \max_{i,j} \nu_j^{(i)}$ is the maximum entry.

2.2 Borodin's product formula

Borodin [**?**] proved:

$$\mathrm{GK}_c(1, q) = \sum_{\pi \in \mathrm{CPP}(c)} q^{|\pi|} = \prod_{s \in \mathcal{C}(c)} \frac{1}{1 - q^{h(s)}}, \tag{6}$$

where $\mathcal{C}(c)$ is the "cylindric shape" determined by the profile c, and $h(s)$ is the hook length of box s in this cylindric shape. This generalizes Stanley's formula (**??**) for RPPs.

The z-refined version involves the transfer matrix (Section **??**).

3 The NILP model: max entry = number of paths

In the non-intersecting lattice path (NILP) interpretation:

> A CPP with max entry $= n$ corresponds to n non-intersecting lattice paths on the cylinder.

Each "layer" (the level set $\{(i,j) : \nu_j^{(i)} \geq k\}$) contributes one path. The paths are non-intersecting because the interlacing conditions force the level sets to be nested.

The maximum entry thus controls the *combinatorial complexity*: $n = 1$ gives a single path (equivalent to a single interlacing chain), while larger n requires coordinating n nested paths on the cylinder.

For the positivity conjecture, the case $n = 1$ is trivial (g_1 counts objects directly), and the difficulty grows with n because the alternating-sign E–H extraction (Section **??**) involves deeper cancellation.

4 The transfer matrix and Schur functions

4.1 Vertex operator formulation

The generating function $\mathrm{GK}_c(z,q)$ can be expressed using vertex operators acting on the ring of symmetric functions. The Pieri rules for Schur functions give:

$$s_\mu[X] \cdot h_r[X] = \sum_{\lambda \in U_r(\mu)} s_\lambda[X], \qquad s_\lambda[X+w] = \sum_{\mu \in D_r(\lambda)} s_\mu[X]\, w^r,$$

where $U_r(\mu)$ (resp. $D_r(\lambda)$) is the set of partitions obtained from μ by adding (resp. removing) a horizontal r-strip.

The coefficient of z^n in $\mathrm{GK}_c(z,q)$ is:

$$g_n(q) = \sum_{\vec{\lambda}} \prod_{i=1}^{k} s_{\lambda^{(i+1)}/\lambda^{(i)}}(1, q, q^2, \ldots), \tag{7}$$

where the sum is over k-tuples of partitions $\vec{\lambda} = (\lambda^{(1)}, \ldots, \lambda^{(k)})$ satisfying the cyclic interlacing (**??**) with all parts $\leq n$.

4.2 Schur positivity

Each factor $s_{\mu/\nu}(1, q, q^2, \ldots)$ in (**??**) is the principal specialization of a skew Schur function, which counts semistandard Young tableaux of skew shape μ/ν by major index:

$$s_{\mu/\nu}(1, q, q^2, \ldots) = \sum_{T \in \mathrm{SSYT}(\mu/\nu)} q^{|T|}.$$

Before specialization, (**??**) is a sum of products of skew Schur functions. Products of skew Schur functions are Schur-positive by the Littlewood–Richardson rule, so:

Proposition 3. *The "unspecialized" transfer matrix expression*

$$\widetilde{g}_n^{\mathrm{TM}}(X) = \sum_{\vec{\lambda}} \prod_{i=1}^{k} s_{\lambda^{(i+1)}/\lambda^{(i)}}(X)$$

is a Schur-positive symmetric function. Under the principal specialization $x_j \mapsto q^{j-1}$, it recovers $g_n(q)$.

Remark 4. *This "unspecialized" lift tracks a* single *alphabet X shared across all k skew Schur factors. It refines the total weight but does* not *distinguish which factor (= which diagonal of the CPP) a box belongs to. The full frequency refinement $\tilde{g}_n(z_1, z_2, \ldots)$ from Section* **??** *is a different (and finer) multivariate lift, and it is* not *symmetric.*

4.3 Connection to SSYT

For RPPs of shape λ (the non-cylindric case), the transfer matrix reduces to:

$$g_{n,\lambda}^{\mathrm{RPP}}(q) = \sum_{\substack{\mu^{(0)}=\emptyset \subseteq \mu^{(1)} \subseteq \cdots \subseteq \mu^{(n)} \\ \mu^{(n)}=\lambda}} \prod_{i=1}^{n} s_{\mu^{(i)}/\mu^{(i-1)}}(1, q, q^2, \ldots).$$

The unspecialized version $\sum \prod_i s_{\mu^{(i)}/\mu^{(i-1)}}(X)$ is a single Schur function evaluated at a product of alphabets: it equals $s_\lambda(X^n)$ under appropriate plethystic substitution.

More concretely, RPPs of shape λ with $\max \leq n$ are in bijection with semistandard Young tableaux of shape λ with entries in $\{1, \ldots, n\}$ (via the level-set/Gelfand–Tsetlin pattern correspondence), giving:

$$\sum_{\substack{\pi \in \mathrm{RPP}(\lambda) \\ \max(\pi) \leq n}} q^{|\pi|} = s_\lambda(1, q, q^2, \ldots, q^{n-1}). \tag{8}$$

The *exact* max-entry count $g_{n,\lambda}(q)$ from (**??**) is then recovered by differencing:

$$g_{n,\lambda}(q) = s_\lambda(1, q, \ldots, q^{n-1}) - s_\lambda(1, q, \ldots, q^{n-2}), \tag{9}$$

the contribution of RPPs whose maximum entry is exactly n. This is the RPP analogue of the E–H extraction: the cumulative $\leq n$ formula is clean (a single Schur specialization), but isolating the $= n$ piece requires a subtraction that destroys manifest positivity.

Remark 5. *For RPPs, positivity of* (**??**) *is immediate:* $g_{n,\lambda}(q) \geq 0$ *because it counts objects. But the analogous extraction for CPPs — Warnaar's* $Q_{n,c}(q)$ *in* (**??**) *— involves the infinite product* $(zq; q)_\infty$ *rather than a simple difference, making positivity a nontrivial conjecture.*

5 Growth diagrams and RSK

5.1 Fomin's growth diagrams

Fomin's theory of growth diagrams [**?**, **?**] provides a unified framework for RSK-type correspondences. A *growth diagram* is a two-dimensional grid of partitions satisfying local rules at each cell.

For an RPP π of shape λ with max entry n:

1. The RPP is encoded as an n-step process, where step k records the k-th level set L_k.
2. Each step corresponds to one "row" of the growth diagram.
3. The local rules (Burge, Knuth, or dual) govern how partitions grow at each cell.
4. The boundary conditions encode the shape λ and the level-set structure.

The key observation:

> *The maximum entry of the RPP equals the number of rows (= insertion steps) in the growth diagram.*

5.2 RSK and the maximum entry

Under the RSK correspondence (in the growth-diagram formulation):

- An RPP π of shape λ with $\max(\pi) = n$ maps to a pair (P, Q) of tableaux.
- The *insertion tableau* P is a semistandard Young tableau with entries in $\{1, \ldots, n\}$. In particular, $\max(P) = \max(\pi) = n$.
- The *recording tableau* Q encodes the order in which cells are filled during the growth process.
- The shape of P (= shape of Q) is determined by the interlacing structure.

Remark 6. *For cylindric partitions, the cyclic boundary condition means the growth diagram "wraps around." This is the setting of the cylindric growth diagrams developed in [**?**, **?**]. The RSK output is no longer a pair of tableaux but rather a pair of* cylindric tableaux*, and the analysis is more delicate.*

5.3 What max entry controls in RSK

The maximum entry n governs three equivalent aspects of the RSK decomposition:

Model	**Max entry** $= n$ **means**
Level sets	n nested layers
NILP	n non-intersecting paths
Growth diagram	n rows (insertion steps)
RSK output	$\max(P) = n$
Schur functions	alphabet of size n

Remark 7 (What max entry does *not* control)**.** *Under RSK, the maximum entry of the input does* not *determine:*

- *The shape of the output tableaux (which depends on the full interlacing structure, not just the depth);*
- *The number of rows or columns of the output (Schensted's theorem relates the longest increasing subsequence to the first row, not to the maximum entry);*
- *Any clean statistic on the* recording *tableau Q.*

*Under the Hillman–Grassl correspondence [**?**] (a different bijection for RPPs), the max entry of the RPP has no known clean characterization on the output side.*

6 Gansner's trace generating function

Gansner [**?**] studied a different multivariate refinement of RPPs: the *trace generating function*, where one tracks the sum of entries along each anti-diagonal.

For an RPP π of shape λ, define the d-th trace as $t_d(\pi) = \sum_{j-i=d} \pi_{ij}$. Gansner showed:

$$\sum_{\pi \in \mathrm{RPP}(\lambda)} \prod_{d} x_d^{t_d(\pi)} = \prod_{u=(i,j)\in\lambda} \frac{1}{1 - x_{j-i}^{h_\lambda(u)}}. \tag{10}$$

Under the specialization $x_d \mapsto q$ for all d, this recovers Stanley's formula (**??**).

Remark 8 (Gansner vs. frequency refinement)**.** *Gansner's trace GF and our frequency GF (Section* **??***) are* different *multivariate refinements:*

	Gansner (traces)	***Frequency***
Variable	x_d *per diagonal* d	z_k *per value* k
Statistic	$t_d = \sum_{j-i=d} \pi_{ij}$	$\#_k = \lvert\{(i,j) : \pi_{ij} = k\}\rvert$
Symmetry	*Symmetric in each* x_d	*Quasi-symmetric in* z_k*'s*
Product formula	*Yes* (**??**)	*No*
Max entry	*Not directly visible*	$\max\{k : \#_k > 0\}$

The frequency refinement can "see" the max entry (it is $\max\{k : z_k$ *appears*$\}$*), while Gansner's trace refinement cannot (the max entry is not a linear function of the diagonal sums).*

7 The E–H factorization and Warnaar's Q polynomials

7.1 Extracting $Q_{n,c}(q)$

The *Euler–Hamburger factorization* separates $\mathrm{GK}_c(z,q)$ into a universal factor and a profile-dependent remainder:

$$\mathrm{GK}_c(z,q) = \frac{1}{(zq;q)_\infty} \cdot F_c(z,q), \qquad F_c(z,q) = (zq;q)_\infty \cdot \mathrm{GK}_c(z,q). \tag{11}$$

Writing $F_c(z,q) = \sum_{n\geq 0} h_n(q) z^n$, the Q polynomials are:

$$Q_{n,c}(q) = (q;q)_n \cdot h_n(q) = (q;q)_n \cdot [z^n]\big((zq;q)_\infty \cdot \mathrm{GK}_c(z,q)\big). \tag{12}$$

Conjecture 9 (Warnaar [**?**])**.** *For all* $n \geq 0$ *and all profiles* c *with* $d = |c| \not\equiv 0 \pmod 3$*, the polynomial* $Q_{n,c}(q)$ *has nonneg integer coefficients.*

7.2 The max-entry structure of Q_n

The extraction (**??**) can be rewritten using the "ground set" formulation:

$$Q_{n,c}(q) = \sum_{j=0}^{n} \sum_{\substack{S\subseteq\{1,\ldots,n\}\\|S|=j}} \sum_{U\subseteq\{1,\ldots,n-j\}} \sum_{\substack{\nu\in\mathrm{CPP}(c)\\\max(\nu)=n-j}} (-1)^{|S|+|U|}\, q^{\sigma(S)+\sigma(U)+|\nu|}. \tag{13}$$

This makes explicit that Q_n involves CPPs at *every* max entry from 0 to n: the j-th term uses CPPs with max $= n - j$. The alternating sign $(-1)^{|S|+|U|}$ must produce miraculous cancellation to yield a nonneg polynomial.

7.3 The q-binomial decomposition

The relationship between Q_n and the "corrected" coefficients $h_m = (q;q)_m \cdot g_m$ is:

$$Q_n = \sum_{j=0}^{n} (-1)^j\, q^{\binom{j+1}{2}} \begin{bmatrix} n \\ j \end{bmatrix} h_{n-j}, \tag{14}$$

with inverse:

$$h_m = \sum_{j=0}^{m} q^j \begin{bmatrix} m \\ j \end{bmatrix} Q_{m-j}. \tag{15}$$

The polynomials $h_m(q)$ are nonneg for all tested profiles (a key finding), and the alternating q-binomial sum (**??**) must preserve nonnegativity — this is the core mystery.

8 Frequency refinement at fixed max entry

8.1 The frequency generating function

For a CPP $\pi = (\nu^{(1)}, \ldots, \nu^{(k)})$, the *value frequency* of k is $\#_k(\pi) = |\{(i,j) : \nu_j^{(i)} = k\}|$. The *frequency generating function* for all CPPs:

$$\widetilde{\mathrm{GK}}_c(\mathbf{z}) = \sum_{\pi\in\mathrm{CPP}(c)} z_1^{\#_1(\pi)} z_2^{\#_2(\pi)} \cdots$$

is quasi-symmetric and L-positive (see [**?**]).

At fixed max entry:

$$\tilde{g}_n = \sum_{\max(\pi)=n} z_1^{\#_1(\pi)} \cdots z_n^{\#_n(\pi)} \;\in\; \mathbb{Z}[z_1, \ldots, z_n].$$

8.2 Partial quasi-symmetry

The function $\tilde{g}_n$ is *not* quasi-symmetric in $z_1, \ldots, z_n$ (the value n is privileged: every CPP with max $= n$ has $\#_n \geq 1$, but may have $\#_1 = 0$). However, it retains a partial quasi-symmetry:

Observation 10 (Computational)**.** *For all tested profiles ($d \leq 5$, $n \leq 3$): writing*

$$\tilde{g}_n = \sum_{m \geq 1} z_n^m \cdot f_{n,m}(z_1, \ldots, z_{n-1}),$$

each slice $f_{n,m}$ is quasi-symmetric in $z_1, \ldots, z_{n-1}$. Equivalently, $\tilde{g}_n \in z_n \cdot \mathrm{QSym}(z_1, \ldots, z_{n-1})[z_n]$.

The argument: relabelling the non-n values $\{v_1 < \cdots < v_k\} \subset \{1, \ldots, n-1\}$ to any other increasing sequence $\{w_1 < \cdots < w_k\} \subset \{1, \ldots, n-1\}$ preserves the interlacing conditions, because:

1. Interlacing among non-n entries depends only on their relative order;
2. Every non-n entry $v < n$, so all inequalities with n-entries are automatically satisfied.

8.3 Failure of slice L-positivity

While the full $\widetilde{\mathrm{GK}}_c$ is L-positive, the individual slices $f_{n,m}$ are *not*. For $n = 3$ (two non-n variables), the M-coefficients satisfy $a_{(k)} > a_{(j,k)}$ for all tested profiles, giving $L_{(a,b)} = M_{(a,b)} - M_{(a+b)} < 0$.

8.4 The variable-count mismatch

The E–H factorization (**??**) couples objects living in different numbers of variables:

- The factor $(q;q)_n = \prod_{k=1}^{n}(1-q^k)$ lifts to the elementary symmetric functions $e_1, e_2, \ldots$ in *infinitely many* variables (or equivalently, to $\prod_{k \geq 1}(1 - z_k)$ in QSym).
- The coefficient $\tilde{g}_n$ is a polynomial in *only n variables* $z_1, \ldots, z_n$.

The product $(q;q)_n \cdot g_n(q)$ in one variable makes sense (multiply two polynomials in q), but any multivariate lift must reconcile infinite-variable and finite-variable objects. This is a structural obstacle to extending the QSym framework from $\widetilde{\mathrm{GK}}_c$ (where it succeeds) to $Q_{n,c}$ (where it fails).

9 Summary: what is and isn't known

Object / Property	**RPPs**	**CPPs**
Max entry = number of layers	Classical	Classical
Hook-product formula ($z = 1$)	Stanley [**?**]	Borodin [**?**]
Max $\leq n$ via Schur functions	Eq. (**??**)	Transfer matrix (**??**)
Growth diagram / RSK	Fomin [**?**]	Cylindric GD [**?**]
Trace GF (Gansner)	[**?**]	Not studied
Frequency GF quasi-symmetric	Yes (P-partition)	Yes [**?**]
Frequency GF L-positive	Expected	Yes [**?**]
Partial QSym at fixed max	Expected	Yes (Obs. **??**)
Slice L-positivity	Unknown	**No** (fails at $n \geq 3$)
E–H factorization	—	Warnaar [**?**]
$Q_{n,c}(q) \geq 0$	—	**Open** (Conj. **??**)

Gaps in the literature:

1. The frequency refinement $\tilde{g}_n$ (at fixed max entry) does not appear to have been studied, for either RPPs or CPPs. The partial quasi-symmetry (Observation **??**) appears new.

2. Gansner's trace generating function has not been extended to cylindric partitions. The cyclic boundary condition complicates the diagonal structure.

3. The Hillman–Grassl image of the maximum entry is unknown. A clean characterization could yield a new approach to the hook-product formula.

4. The failure of slice L-positivity (despite aggregate L-positivity) has not been observed before and may constrain which algebraic approaches can prove positivity of $Q_{n,c}$.

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